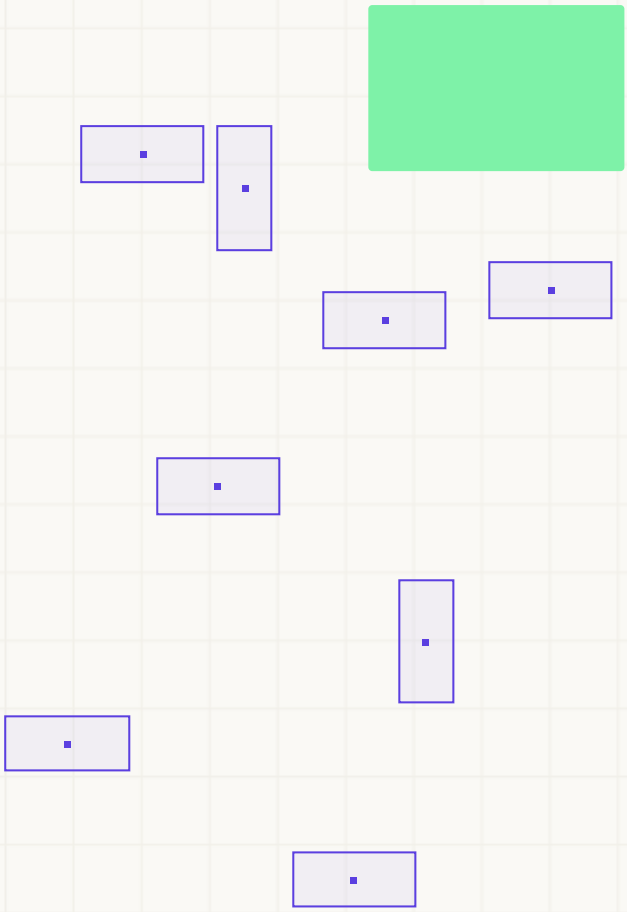
LAAI | AI FOR CONSTRUCTION



AI terms explained: a glossary for construction

Tokens, models, context windows, GPUs, RAG, agents and more: the words you hear in every AI conversation, explained in plain language with examples from architecture and building.

INSIDE

A one-page picture of how AI works and 76 terms in 8 groups

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THE WHOLE PICTURE FIRST

What happens when you ask an AI a question

Most AI words describe one step of this simple chain. Once you see it, the rest of the glossary falls into place.

1 · PROMPT

You ask

A question, an instruction and maybe a file, such as a tender PDF.

→

2 · TOKENS

Text is cut up

Everything is split into small pieces of words, called tokens.

→

3 · INFERENCE

The model predicts

On GPUs in a data centre, the model predicts the most likely next token, one at a time.

→

4 · OUTPUT

You get an answer

The tokens are joined back into text, code or an image.

Context window everything the model can see at once: `instructions` `your files` `the conversation so far` `its answer`

What a token looks like

`Sum` `mar` `ise` `this` `tender` `in` `five` `points`

An approximate split: each colour is one token. In English, 1 token is about ¾ of a word, so 100 pages of text are roughly 60,000–70,000 tokens. Spanish and other languages usually need a few more tokens per word.

Training: once, by the vendor

The model learns patterns from huge amounts of text, images and code. It takes months and thousands of GPUs, and ends at a knowledge cutoff date.

Inference: every time you ask

Using the trained model to answer. This is what you pay for per token, and it does not change the model unless the vendor trains on your data.

Three things people mix up

- ☐ **The app** is what you use: the Claude, ChatGPT or Gemini website, or Copilot inside Word.
- ☐ **The model** is the engine inside the app. Vendors release new versions several times a year.
- ☐ **The API** is the door for other software, such as your own scripts, to use the same model.

THE BIG WORDS

AI, models and what is inside them

These are the words used in headlines. Many are nested: a large language model is a kind of generative AI, which is a kind of machine learning.

Artificial intelligence (AI)

Software that does tasks that normally need human intelligence, such as understanding text, recognising images or making predictions.

EXAMPLE Reading a specification and listing the fire requirements.

Machine learning

AI that learns patterns from examples instead of following rules written by a programmer.

EXAMPLE A model that learns to spot cracks from thousands of facade photos.

Generative AI GENAI

AI that creates new content: text, images, code, audio or 3D, instead of only classifying or predicting.

EXAMPLE Drafting a method statement or a render from a massing model.

Model

The trained “brain” that does the work. Products such as Claude, ChatGPT or Gemini are apps built on top of one or more models.

EXAMPLE One app can offer a fast, cheap model and a slower, smarter one.

Large language model (LLM)

A model trained on huge amounts of text to understand and write language. The core of today’s AI assistants.

Neural network

The structure inside a model: layers of simple maths units connected together, loosely inspired by the brain.

Parameters WEIGHTS

The numbers a model learns during training. Large models have billions of them; more is not always better for your task.

Training data

The text, images and code a model learned from. It explains what the model knows, and also its gaps and biases.

EXAMPLE A model may know Eurocodes better than a local building ordinance.

Knowledge cutoff

The date the training data ends. The model does not know later events unless it can search the web or read your files.

EXAMPLE A new regulation published after the cutoff.

Multimodal

A model that works with more than text: images, PDFs, drawings, audio or video.

EXAMPLE Uploading a plan photo and asking what is missing.

PROMPTS AND ANSWERS

The words for what goes in and what comes out

These terms explain why the same tool gives a great answer to one person and a poor one to another.

Prompt

What you write to the AI: the question, the instructions and any context. Clear prompts give better results.

EXAMPLE "You are a site manager. Summarise this tender in 5 points for the client."

System prompt

Standing instructions set before the conversation, by the app or by you, that shape every answer.

EXAMPLE "Always answer in Spanish and use CTE terminology."

Token

A small piece of text, often part of a word. Models read, write, limit and charge by tokens.

EXAMPLE 1,000 tokens ≈ 750 English words.

Context window

The maximum number of tokens the model can consider at once. In 2026 it ranges from about 100,000 to over a million tokens.

EXAMPLE A large window can hold a full tender and its annexes.

Hallucination

A confident answer that is wrong or made up, such as a standard clause or a reference that does not exist.

EXAMPLE Always check numbers, standards and citations.

Grounding

Making the model answer from specific sources you give it, and cite them, to reduce hallucinations.

EXAMPLE "Answer only from the attached specification and quote the page."

Reasoning model THINKING

A model that works through a problem step by step before answering. Slower and uses more tokens, but better for complex tasks.

EXAMPLE Checking a structural calculation logic or debugging a script.

Temperature

A setting for how predictable or creative the answers are. Low for facts and code, higher for ideas.

Prompt library

A shared collection of prompts that work well for your company's typical tasks.

EXAMPLE One tested prompt for tender summaries, one for RFIs.

Chatbot vs assistant

A chatbot answers questions; an assistant can also read your files, use tools and carry out tasks.

HOW AI IS PRICED

What you actually pay for

There are two ways to pay: a subscription per person, or pay per token through an API. Knowing the difference avoids surprises.

Subscription (seat)

A fixed price per person per month for an app. Business plans add admin controls and data protection terms.

EXAMPLE A team plan for 25 people.

API APPLICATION PROGRAMMING INTERFACE

A way for software to talk to a model directly. Used by developers and custom tools, and billed per token.

EXAMPLE A Rhino script that sends a model's data to an AI and gets a report back.

Input and output tokens INPUT / OUTPUT

Input is what you send (prompt and files); output is what the model writes. Output usually costs several times more per token.

Price per million tokens

How API prices are quoted. Small models cost cents; the most capable ones cost more, but a typical document task still costs little.

EXAMPLE Summarising a 100-page tender, about 70,000 tokens, usually costs well under €1.

API key

A secret password that lets a tool use your paid API account. Never share it or put it in a file others can see.

Rate limit

The maximum number of requests or tokens you can use per minute or per day on a plan or API.

EXAMPLE A heavy user hits the daily limit of a basic plan.

Caching PROMPT CACHING

Reusing the same long input, such as a standard, across many questions, so it costs less and answers faster.

Free plans FREE TIER

Useful to try things, but they have lower limits and often allow your data to be used for training. Not for client work.

```
Worked example: summarising a 100-page tender through an API
70,000 input tokens × €3 / million + 1,500 output tokens × €15 / million
≈ €0.23
```

Illustrative prices for a capable model; check each vendor's current price list. For most offices a subscription per person is simpler; the API matters when you build your own tools.

WHERE AI RUNS

Chips, data centres, cloud and local models

The physical side of AI explains why it costs what it costs, where your data travels, and when a local model makes sense.

GPU GRAPHICS PROCESSING UNIT

The chip that runs AI. Originally made for graphics; it does thousands of small calculations at once.

EXAMPLE The same kind of chip that speeds up rendering in Enscape or Twinmotion.

Data centre DATA CENTER

A building full of servers and GPUs where models are trained and run. It uses a lot of electricity and cooling.

EXAMPLE Data centres are now a major construction market in their own right.

Compute

The amount of processing power used to train or run AI. More compute usually means more cost and energy.

Cloud

Using AI on the vendor's servers over the internet. Easy and powerful; your data leaves your office under the vendor's terms.

Local or on-premise model

A model that runs on your own computer or server. Data stays inside, but the models are smaller and need a strong GPU.

EXAMPLE Summarising confidential documents without sending them out.

Open-weight vs closed model OPEN WEIGHTS

Open-weight models can be downloaded and run yourself; closed models are only used through the vendor's app or API.

Data residency

Where your data is stored and processed. Some business plans let you keep it in the EU, which helps with GDPR.

Latency

How long you wait for the answer to start. Matters for live tools inside Rhino or Revit.

Energy use

Training and running AI uses significant electricity and water for cooling. Smaller models and fewer wasted requests reduce it.

Edge AI

AI that runs directly on a device, such as a phone, a scanner or a site camera, without the cloud.

EXAMPLE A helmet camera that detects missing safety gear.

MAKING AI KNOW YOUR COMPANY

How AI uses your own documents and software

A general model does not know your projects, standards or prices. These are the ways to connect it to them, from simplest to most complex.

Uploading files

The simplest way: attach the documents to the conversation so they are in the context window.

Projects and knowledge bases

A space in an AI app where you store documents and instructions once and reuse them in every chat.

EXAMPLE A project with your detail library and drawing standards.

RAG RETRIEVAL-AUGMENTED GENERATION

The AI first searches your documents for the relevant parts, then answers using them. Used for large archives that don't fit in the context.

EXAMPLE Asking questions across 10 years of project files.

Embeddings

Text turned into lists of numbers that capture meaning, so a computer can find similar content even with different words.

EXAMPLE "Fire door" also finds "EI2 60 door".

Vector database VECTOR DB

A database that stores embeddings and finds the closest matches quickly. The search engine behind most RAG systems.

Fine-tuning

Training an existing model a bit more on your own examples. Rarely needed now; good prompts and RAG usually work better.

Connectors INTEGRATIONS

Links that let an AI app read from other tools, such as email, Drive, SharePoint or a project platform.

MCP MODEL CONTEXT PROTOCOL

An open standard, introduced by Anthropic in 2024 and now widely supported, for connecting AI assistants to software and data.

EXAMPLE RhinoMCP lets an AI assistant work directly inside Rhino.

Training on your data OPT-OUT

Whether the vendor may use your conversations to improve its models. Business plans usually exclude it; check the terms.

Data retention

How long the vendor keeps your prompts and files. Some offer zero data retention for sensitive work.

WHEN AI DOES THE WORK

From answering to acting

The newest AI tools don't just answer: they run steps, use software and write code. This is where the biggest time savings in construction are today.

AI agent

An AI that plans and carries out several steps towards a goal, using tools, checking results and trying again.

EXAMPLE "Check every sheet in this model for missing room names and fix them."

Tool use FUNCTION CALLING

When a model calls other software, such as a calculator, a web search, Excel or Rhino, instead of only writing text.

Copilot

A general name for an AI assistant built into the software you already use. Also the name of Microsoft's and GitHub's products.

AI coding assistant

An AI that writes, explains and fixes code with you. It lets people who are not programmers build useful scripts.

EXAMPLE Claude Code, GitHub Copilot.

Script

A short program that automates a task inside software. In construction, usually Python or C# for Rhino, Grasshopper, Revit or Dynamo.

EXAMPLE A script that makes area schedules for every apartment.

Workflow automation

Connecting steps and tools so a process runs by itself, with AI handling the parts that need reading or writing.

EXAMPLE New tender in the inbox → summary → task for the team.

Human in the loop

A person reviews or approves the AI's work at key steps. Essential for anything that leaves the office.

Vibe coding

Building software by describing what you want and letting AI write the code. Fast for prototypes, but the result still needs testing.

FOUR LEVELS, FROM CHAT TO AGENT

01
Chat
You ask, it answers. You copy the result into your work.

02
Your files
It reads your documents and answers from them.

03
Tools
It uses software for you: Excel, Rhino, a web search.

04
Agent
It plans and does several steps; a person reviews the result.

WORDS FROM CONTRACTS AND POLICIES

The terms behind safe AI use

You will meet these in vendor terms, client contracts and the EU AI Act. Our guide on AI risks explains each risk in detail.

EU AI Act

The EU law on AI. It sorts uses by risk, bans some, sets strict duties for high-risk uses and transparency duties for others.

AI literacy

Since February 2025 the AI Act asks organisations that use AI to make sure their staff understand it well enough.

EXAMPLE Training your team is part of meeting this.

High-risk AI

Uses with strict duties under the AI Act, such as AI in hiring, education assessment or safety components of products.

GDPR

The EU data protection law. It applies whenever personal data, such as names, emails or site photos of people, goes into an AI tool.

Shadow AI

Staff using AI tools the company has not approved, often personal accounts with client data.

Bias

Systematic errors from the training data, such as favouring one region's standards or one type of building.

Prompt injection

Hidden instructions inside a document or web page that try to make the AI do something you did not ask.

EXAMPLE White text in a tender PDF telling the AI to rate a bid highly.

Copyright and ownership

Who owns AI-made content and whether it copies others' work. Agree it in contracts, especially for designs and images.

Deepfake

AI-made images, audio or video that look real. The AI Act requires labelling them; AI renders should be marked as concepts.

AI policy

A short internal document saying which tools are approved, what data may go where, and who checks the output.

WHERE AI MEETS THE BUILDING

Terms you will hear in AEC

Some of these are older than generative AI, but they now come up in almost every conversation about AI in architecture, engineering and construction.

Computer vision

AI that understands images and video: detecting elements on drawings, defects in photos or progress on site.

OCR OPTICAL CHARACTER RECOGNITION

Turning text in scans and images into editable text. The first step for reading old drawings and archive documents.

AI takeoff

Measuring areas, lengths and counts automatically from PDF drawings or models.

Generative design

Software that creates many design options from goals and rules, so you can compare them.

EXAMPLE Testing 200 site layouts for unit count and daylight.

Parametric design

A model driven by parameters and rules, often in Grasshopper or Dynamo. Not AI itself, but AI now helps write and edit it.

Point cloud

Millions of 3D points from a laser scan or photogrammetry. AI helps classify them into walls, floors and elements.

EXAMPLE Scan-to-BIM of an existing building.

Digital twin

A live digital model of a building, connected to real data from sensors, used to run and maintain it.

Model checking

Rules or AI that check a BIM model for errors, missing data or clashes before it is shared or produced.

IFC INDUSTRY FOUNDATION CLASSES

The open file format for BIM. Clean IFC data makes AI tools work much better across different software.

Text-to-image

Creating an image from a written description, or from a sketch or massing plus a description.

NEXT STEP

Words are the start. Using AI well is the next step.

I'm Lavinia Lorena Jilavu, a BIM expert in Málaga. I helped bring AI into daily work at a modular timber company and built around 30 tools with it. I can help you:

- ☐ **Train your team** so everyone understands these terms and uses AI well, in Spanish, English or Italian.
- ☐ **Find where AI saves you the most hours**, in a workflow audit.
- ☐ **Build custom tools** for Rhino, Grasshopper, Revit or Excel, yours to keep.
- ☐ **Read the other free guides**: getting started, AI risks, AI tools and AI strategy.

Book a free first session

Half a day on one of your real tasks, with no commitment.

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